

Operation manual for PA-ARH-100D CF34-mount compact 30kV RHEED system with double-stage differential pumping



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1 Introduction

1-1 Outline of Pascal RHEED system

Reflection high-energy electron diffraction (RHEED) is a useful tool for a surface analysis *in-situ* in PLD or other thin-film deposition process.

Pascal RHEED is a compact system for easy installation and operation. The RHEED system consists of electron-gun with 30kV off acceleration voltage, high-voltage power supply with remote controller, and fluorescent screen.

Units or parts of the RHEED system are danger. See following chapters and individual documents for assembled parts which show functions of the units and the parts in detail.



1-2 Caution



X-ray is generated when an electron beam is irradiated to a substrate with RHEED (Reflection High Energy Electron Diffraction) gun. While irradiating, be away from other viewing ports than the screen.



Wiring relay terminals and wire connectors wire numerous electric power lines and sensor signal cables. In the case that their covers are removed or the connectors are separated, be sure that the turning the control power supply off.



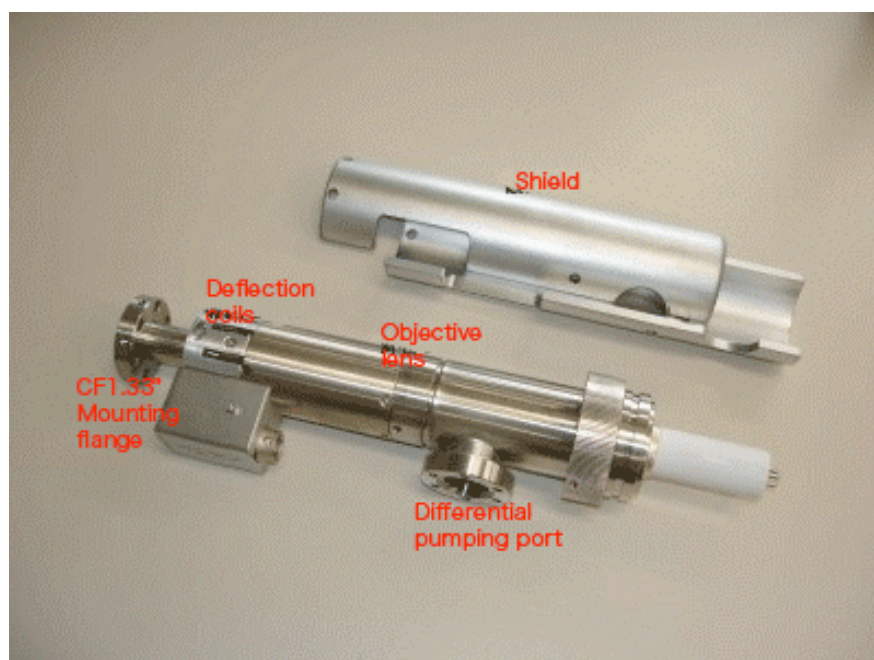
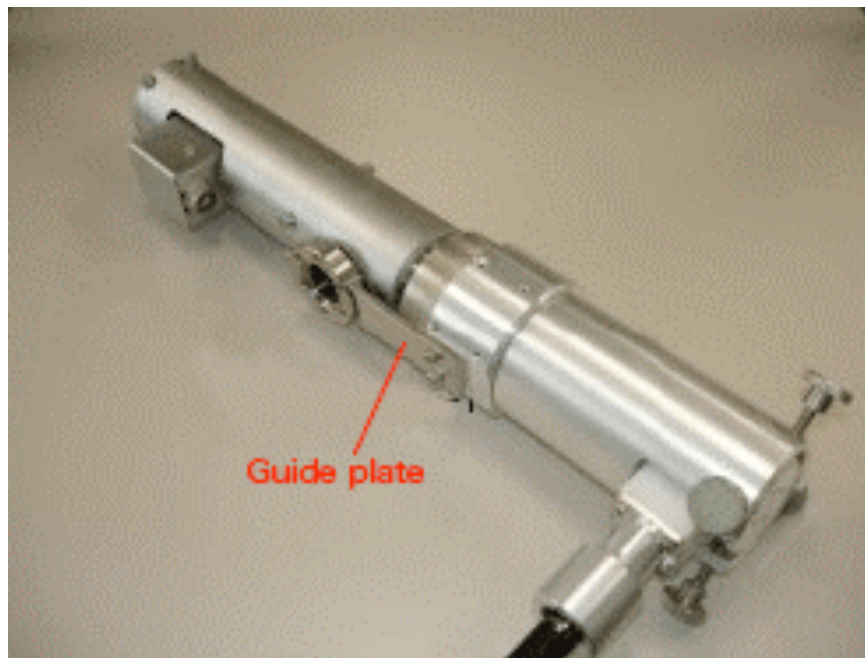
This system basically requires being electrified continuously. Connecting the system to a power line of insufficient capacity may cause an accidental electric power failure or system malfunction. The rating of electric power line should be sufficient.

2 Construction of RHEED system

2-1 Electron-gun

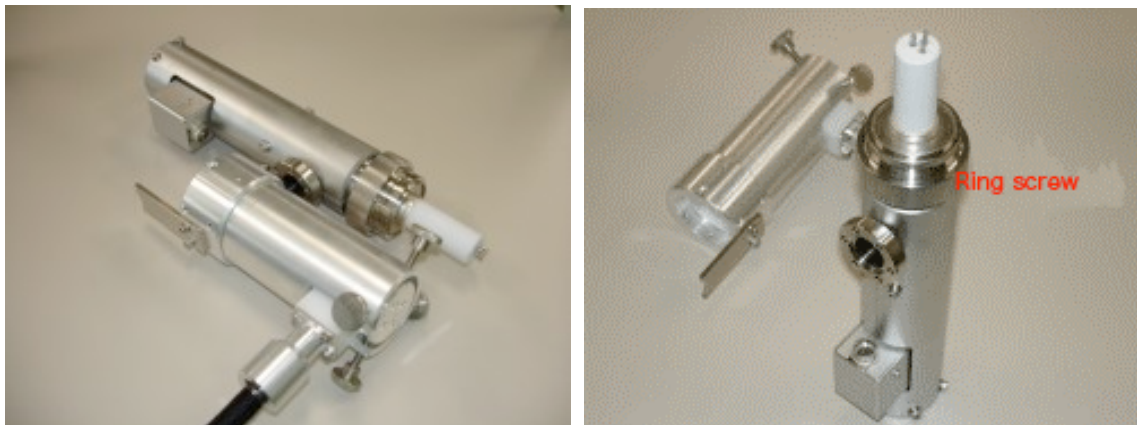
(1) Main body with mounting flange

The electron-gun body is parts for electron emission. The body also has one mounting flange to a chamber, a high-voltage power supply connector, a control cable connector for electron deflection, and a differential pumping port. Mounting flange of the e-gun body is CF1.33" (ICF34). The body is covered with shield plates.



(2) High-voltage power supply connector

This part feeds high-voltage power from the power supply to the electron-gun. Inside the connector three pins sticks out an insulator with a connecting direction.



(3) Electron-gun adjuster knob

There are four adjuster knobs at rear end of the electron-gun. They change a direction of the Wehnelt assembly towards the anode aperture.



(4) Wehnelt assembly

The part is a combination of a Wehnelt electrode and its base to fix. The Wehnelt electrode convergences the electron beam with appropriate bias voltage. A small hole of the Wehnelt electrode is the collimation point. A cusp of a hairpin filament is set at 1 mm inside the Wehnelt electrode.

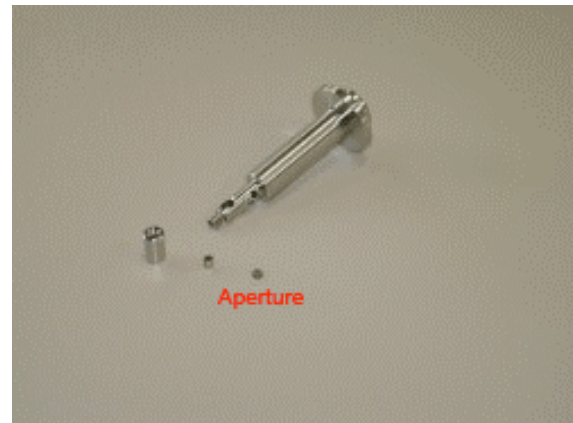
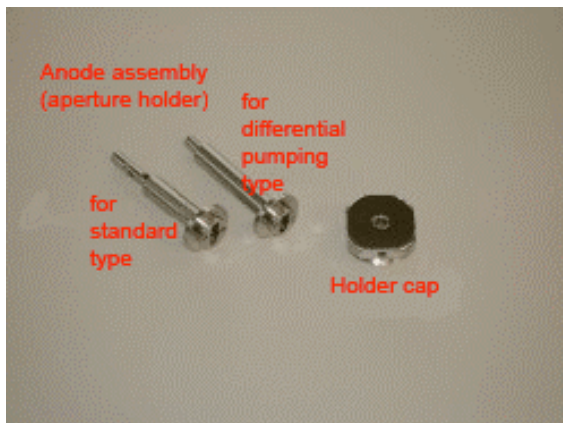


(5) Filament assembly

The part is a combination of a hairpin shaped tungsten filament and its base to fix. Mounting position of the filament can be changed by fixing screws at side of the assembly part.

(6) Anode assembly

The part consists of an (objective) aperture and its holder. The aperture is set at front side of the holder. The aperture has a 0.2 mm diameter hole to convergence the electron beam.



(7) Objective lens

This part works focusing of the electron beam at distance more than 200 mm from the mounting flange of the electron-gun to the fluorescent screen. Focus of the electron-beam is controlled by the remote controller.

(8) Deflection coils

Two set of coils can change the electron beam direction at X (horizontal) or Y (vertical). Changing the electron beam direction is operated by the remote controller. Signal for focusing and deflection of the electron beam is bundled with one cable and connected to the electron-gun terminal near the differential pumping port. Definition X and Y corresponds the direction of the electron-gun body at which the differential pumping port goes down.

(9) Differential pumping port

This is a port to connect a pump when the electron-gun is driven at higher process pressure. The upper threshold pressure near the filament (inside the anode aperture) is around 10-2 Pa. Factory

recommends a turbo molecular pump with pumping speed more than 70 L/s and a supplemental pump like a rotary pump.

(10) Magnetic shield plates

This part covers the electron-gun body to prevent external magnetic-field fluctuation.

2-2 High voltage power supply

(1) Main body

This unit feeds high-voltage power to drive the electron gun. Another function is control of some coils for electron-beam focusing or deflection. The case of the unit can be set onto a JIS conformity rack. The front panel of the unit has a main power switch and a connector for remote controller. The rear panel of the unit has a high-voltage power cable terminal, a coil control terminal, two ground terminals, and a power plug.

(2) Remote controller

This unit has a function to change the electron-gun driving parameter, for example, acceleration voltage, cathode current, and coil current. The top panel of the remote controller has an indicator, a selector, five switches, and five adjuster dials. The five switches and the adjuster dials are for HV (acceleration voltage), FIL (cathode current), FOC (electron-beam focusing), X, and Y (electron-beam deflection), respectively.



2-3 Fluorescent screen

A fluorescent screen is provided in accordance with the port size of the chamber. It consists of two parts whenever a user chooses any screen size.

(1) Screen plate

It is a glass plate with an appropriate diameter in conformity to a port of CF6" or CF8". Conductive

thin film “metal-back” is coated on the glass plate before phosphor coating. The phosphor is coated by water-emulsion base glass.

(2) Mounter

It is a metal band with small screws to fix the screen plate to the port.

3 Installation

3-1 Mounting electron-gun to a chamber

- (1) Mount the electron-gun to a chamber through CF1.33” flange.
- (2) Set turbo molecular pump(s) (TMP(s)) when the RHEED unit needs it (them) for differential pumping. The setting depends on a chamber layout. In usual case the TMP(s) is/are settled on a chamber lack.
- (3) Connect flexible pipe(s) from the electron-gun to the TMP(s).

3-2 Setting HV power supply

- (1) Confirm that the HV power supply unit is settled on a lack.
- (2) Connect the remote controller cable.
- (3) Connect AC100V power inlet cable.

3-3 Fixing fluorescent screen

(pending)

3-4 Cable setting between RHEED e-gun and HV power supply

- (1) HV cable: Notice pin layout of the connector or plug on rear side of the HV power supply.
- (2) HV GVD cable: It is bundled with HV cable.
- (3) GND cable: It connects the electron-gun body and rear panel of the HV power supply.
- (4) Lens and coil control cable: It connects the electron-gun body and rear panel of the HV power supply for electron-beam focusing and deflection. This cable cannot be baked. So please disconnect the cable when chamber baking.

4 Operation guide

4-1 Initial setting

- (1) Adjust HV, FIL, or FOCUS value to zero in the remote controller.
- (2) Adjust DEF-X or DEF-Y value to 5.00 in the remote controller.
- (3) Choose display mode to HV or FIL by selector button to observe the filament current first.
- (4) Push FIL button to turn on the filament. Check that the filament current runs around 500 mA after operation.

An interlock function works not to turn on the filament when FIL value is not zero first.

- (5) Push HV button to turn on the acceleration.

An interlock function works not to turn on the acceleration unless the filament current runs. This function effectively works when one mistake the acceleration operation under HV cable disconnection from the electron-gun.

4-2 HV drive for electron beam emission

- (1) Choose display mode to HV by selector button.
- (2) Raise HV value to an appropriate acceleration voltage by HV adjuster. The RHEED e-gun takes warm-up time in a few minutes because of thermal electron emission. So it is necessary to adjust the HV value with considering about 10% HV drop in the warm-up time.

Extraordinary discharge probably occurs when the acceleration operation at initial use of the electron gun or just after differential pumping. Protection function set HV value to zero when discharge current exceeds 250 μ A. Please restart with setting zero values of FIL and HV.

4-3 Filament drive and initial electron-beam alignment

- (1) Choose display mode to FIL by selector button.
- (2) Raise FIL value to the appropriate filament current by FIL adjuster. Electron-beam usually emits when the filament current reaches around 2000 mA.
- (3) Check electron-beam current after the electron-beam emission. The electron-beam current is indicated on the lower line of the display when choosing HV display mode.

First the electron-beam current rapidly increases and then slowly increases. Gradually increase of the electron-beam current is caused by space-charge limited law. And it is the stable operation condition. But less electron-beam current (and the filament current) brings longer filament lifetime.

The electron-beam current occurs to show extraordinary value when the filament is properly set in the Wehnelt-filament assembly. First case is larger current over than 250 μ A. The other case is small current near zero. Reassemble with adjusting filament position when one sees above cases. The electron beam current occurs to show zero due to break of the filament. Exchange the

filament in the case.

Extraordinary discharge occurs and makes HV to turn off when the filament unintentionally contacts the Wehnelt. Inform PASCAL about the accident in this case.

- (4) Maximize the electron-beam spot on the fluorescent screen by electron-gun adjuster knob.

Remove obstacles and eliminate external magnetic-field disturbances.

Use the electron-gun adjuster knobs to bring back the electron-beam spot onto the fluorescent screen.



4-4 Focus tuning

- (1) Choose display mode to FOC by selector button.
- (2) Minimize the electron-beam spot on the fluorescent screen by FOCUS adjuster.

4-5 X-Y Deflection setting

- (1) Choose display mode to X (or Y) by selector button.
- (2) Set to an appropriate horizontal (or vertical) position on the fluorescent screen by DEF-X (or DEF-Y) adjuster.

Check the DEF-Y parameter after setting because the vertical position of the electron-beam is important to make the electron-beam path including reflection on the substrate surface.

4-6 Shutdown

- (1) Return the all values to initial state as follows in order.
 - a) DEF-X, DEF-Y: 5 in the adjusters. It is not necessary to return if one wants to restart the RHEED system as is.
 - b) FOCUS: 0 in the adjuster. It is not necessary to return if one wants to restart the RHEED system as is.
 - c) FIL: 0 in the adjuster.
 - d) HV: 0 in the adjuster.
- (2) Turn off all function buttons in order: DEF-X, DEF-Y, FOCUS, HV, and FIL.

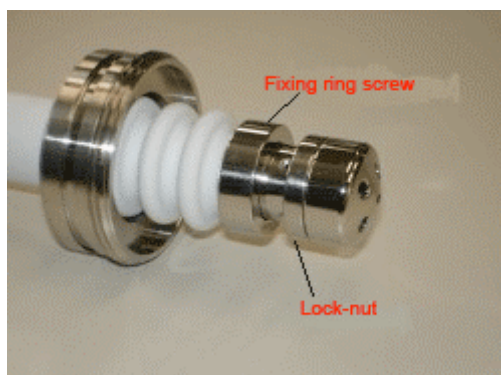
5 Maintenance

5-1 Filament exchange

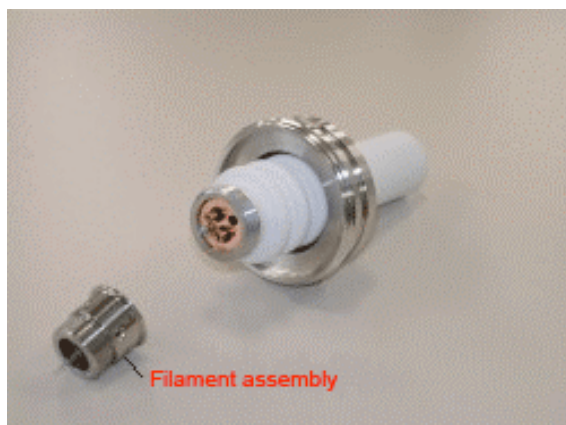
- (1) Disconnect the HV cable connector.
- (2) Take the filament / Wehnelt assembly off through the mounting flange.



- (3) Loose the lock-nut of the Wehnelt cap.
- (4) Remove the Wehnelt fixing ring without rotation of the Wehnelt body.



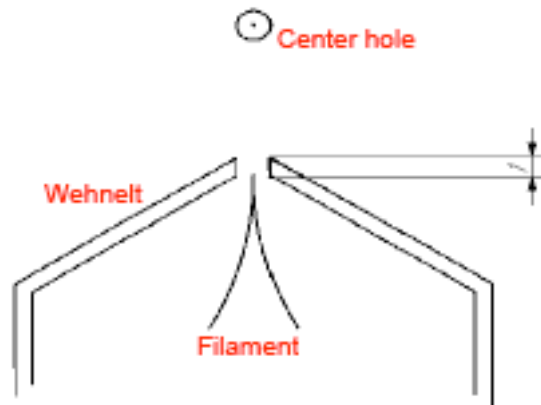
- (5) Loose the filament fixing screws.
- (6) Pick up the old filament and insert new filament in direction as two pins go into plugs.
- (7) Rebuild the filament / Wehnelt assembly.



- (8) Adjust filament fixing position by the fixing screws during rebuilding the filament assembly. Proper fixing position is shown in below figure.

(Confirm that the tip point of the hair-pin shaped filament is at center of the small hole in top view of the Wehnelt. And confirm that the tip point of the filament goes 1 mm down from top surface of the Wehnelt in side view or in slant view by naked eye.)

Do not touch the filament during adjusting, or the filament is broken.



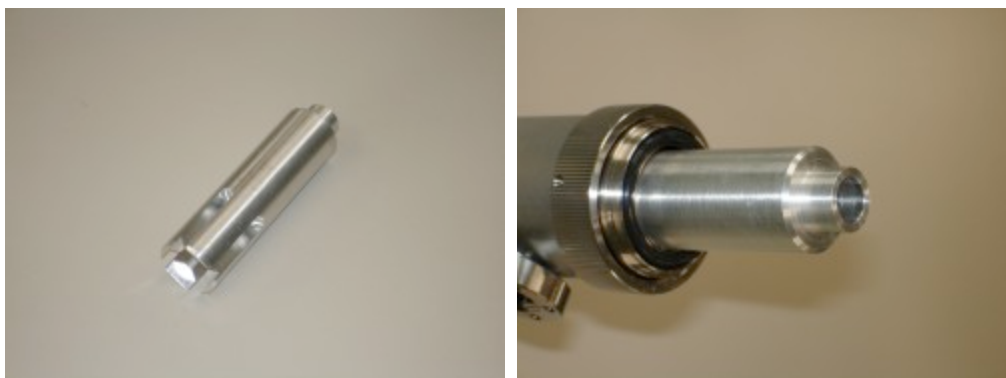
- (9) Fasten the filament fixing ring.
 (10) Fasten the Wehnelt fixing ring without rotation of the Wehnelt body.
 (11) Mount the filament / Wehnelt assembly to the e-gun body with setting O-ring.

5-2 Aperture exchange

- (1) Disconnect the HV cable connector.
 (2) Take the filament / Wehnelt assembly off through the mounting flange.
 (3) Remove the cap screw to fix the aperture holder inside the electron-gun body.



- (4) Insert the mounter / remover tool to remove the aperture holder.
The aperture holder has screw. So the aperture holder can be removed by counterclockwise rotation.



- (5) Remove the aperture holder by tweezers.

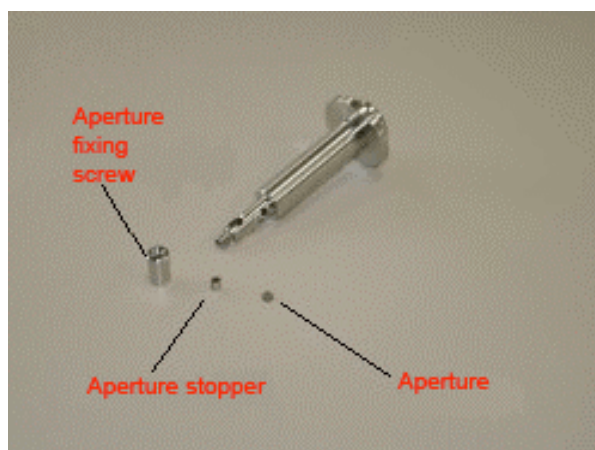


- (6) Pick up the old aperture.

The aperture is set at chamber side of the aperture holder with a stopper and a fixing screw.

- (7) Insert new aperture.

Check that the aperture is clean without dusts or stains.



- (8) Rebuild the aperture holder and the filament / Wehnelt assembly.

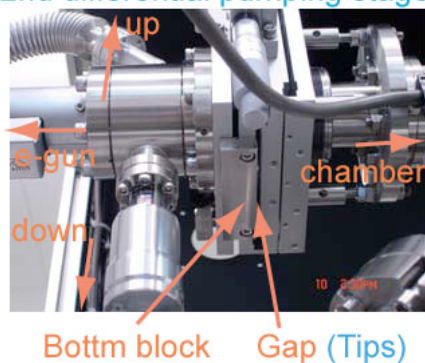
5-3 Cleaning

Cleaning of the Wehnelt or the aperture is necessary when this parts is contaminated by impurities from deposition process or filament material due to unintentional evaporation. Main procedure of the cleaning is polishing by polisher powder.

without e-beam focusing (or with defocusing).

- (4) Record the focus value on the e-gun remote controller. Next set the focus value to zero again.
- (5) Choose the second aperture with ϕ 0.5 mm, smaller diameter than that at initial setting.
- (6) Make the e-beam path through the second aperture with adjusting the XY-axis deflection and the focusing of the e-beam in remote controller.
- (7) Move the tilt table down as the table contacts to the bottom block (slant position). Loose and fix the screw before and after the tilt setting, respectively.

Bottom view of
2nd differential pumping stage



Tips to set “slant position” of the second differential pumping stage

Close the gap near the bottom block.

Notice that the e-beam guide pipe in the chamber touches the chamber wall or units.

- (8) Make the e-beam path as the e-beam irradiates the sample (substrate) surface. Next adjust the screw until one can see the proper diffraction pattern on the fluorescent screen.
- (9) Move back the tilt table up to the “parallel position”. Next check that the direct-beam spot appears on the screen.
- (10) Observation of the diffraction pattern and direct-beam spot can be seen after the e-beam alignment at “slant position” and “parallel position”, respectively.
- (11) The e-beam path fluctuates when the e-gun driving condition, for example the acceleration voltage or the filament current, changes.

In normal operation it is better that the e-beam alignment is done with the ϕ 3 mm aperture and then get the aligned e-beam with ϕ 0.5 mm aperture before RHEED observation at “parallel position”.

6-4 Combination of aperture diameters and RHEED driving pressures

The table show the reference data of upper limit of RHEED driving pressure when the aperture with each diameter is used. The values have deviations upon the vacuum system or process-gas flow-rate.

Aperture diameter size	ϕ 0.5 mm	ϕ 1.0 mm	ϕ 3.0 mm
Capable RHEED e-gun driven pressure	3.0+2 Pa	3.0+1 Pa	3.0+0 Pa

Condition: TMPs are used with 250 L/s at first pumping port and with 50 L/s at second pumping port.

Operation manual for CF34-mount compact RHEED system with
second differential pumping stage

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